

Donghyun Park

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RESEARCH INTERESTS

Sequential Inference (E-values and Confidence Sequences), Boundary-Crossing Inequalities, Concentration Inequalities, and Bandit Problems

EDUCATION

Seoul National University (SNU)

Mar. 2021 – Aug. 2027 (Expected)

B.S. in Statistics

Double Majors: Mathematical Sciences; Computer Science and Engineering

- Cumulative GPA: 4.13/4.30, Statistics GPA: 4.30/4.30, Mathematics GPA: 4.25/4.30
- Left for Mandatory Military service: *Jul. 2023 – Apr. 2025*

PUBLICATIONS

- [1] Donghyun Park, Junhyun An, Taehyoung Kim, and Jisu Kim. *Subsampling Confidence Bound for Persistent Diagram via Time-delay Embedding*. 2026. arXiv: [2512.06324](https://arxiv.org/abs/2512.06324) [math.ST]. URL: <https://arxiv.org/abs/2512.06324>.

RESEARCH EXPERIENCE

High-Dimensional Multiple Testing Lab

Feb. 2026 – Present

Undergraduate Research Intern — Advisor: Junyong Park, Dept. of Statistics, SNU

Will be edited.

Growth-Rate-Optimal E-Variables for parametric testing [Research Note]

- Reviewed growth-rate-optimal e-variables and reverse information projections for hypothesis testing in exponential families.
- Derived growth-rate-optimal e-variables for general variance testing in multivariate normal models.
- Investigated extensions to exponential-family testing from an information-geometric perspective, with a focus on how mixture constructions smooth sharp geometric curvature.
- Supported by LX Science Fellowship. (Research fund of \$1000)

Bio and Health Informatics Lab

Sep. 2026 – Present

Undergraduate Research Intern — Advisor: Sun Kim, Dept. of Computer Science and Engineering, SNU

Will be edited.

Statistics and Artificial Intelligence with Topology Lab

Apr. 2025 – Jan. 2026

Undergraduate Research Intern — Advisor: Jisu Kim, Dept. of Statistics, SNU

Topological Periodicity Test via Confidence Bound of Time Delay Embeddings [1]

Joint work with two undergraduate colleagues

- Developed a geometry-based periodicity test using time-delay embeddings as an alternative to conventional Fourier-based methods.
- Established a quantitative lower bound on the reach of time-delay-embedded point clouds and used this geometric property to construct a test with controlled type-I error.
- Evaluated the proposed method through simulations and real-data analyses, comparing its performance with existing Fourier-based periodicity tests.
- Supported by Student-Directed Education Program from SNU. (Research fund of \$500)

PRESENTATIONS

Recent Developments in Sequential Testing and E-Processes, *Workshop Presentation, 2026 Workshop on Robust Inference for High-Dimensional Complex Data, SNU, Aug. 2026*, [\[Slides\]](#)

Subsampling Confidence Bound for Persistence Diagrams via Time-Delay Embedding, *Statistics and Artificial Intelligence with Topology Lab Weekly Seminar, SNU, Mar. 2026*, [\[Slides\]](#)

HONORS AND AWARDS

National Science & Technology Scholarship *Mar. 2021 – Dec. 2026*

Full-tuition scholarship for eight semesters of undergraduate study

Best Research Award, 1st Place, Student-Directed Education Program *Jan. 2026*

Gold Prize, Korean Undergraduate Mathematics Competition *Dec. 2025*

3rd Place, Simon Marais Mathematics Competition, East Division *Dec. 2025*

Dean's List, SNU College of Natural Sciences (6 semesters) *2022 – 2026*

TEACHING EXPERIENCE

Peer Tutor, Regression Analysis and Practice course *Mar. 2026 – Jun. 2026*

- Mentored five peers in understanding theoretical concepts and constructing rigorous proofs.

Lecturer, An Invitation to the p-adic numbers *Dec. 2025 – Jan. 2026*

- Designed and delivered a four-week intensive lecture series for six undergraduate peers.
- Covered analysis and algebra over the p-adic numbers, Hensel's lemma, and Gauss's three-square theorem. [\[Lecture Notes\]](#)

EXTRACURRICULAR ACTIVITIES

Meari, School band of SNU *2021 – 2023*

SKILLS

Programming Language	R, C/C++, Python
Language	Korean (native), English (advanced)